

Practitioner perspectives on institutional collaboration and multilateral climate funds in the Group of 77 and China

Ravindu Nawanjana

Guildhall School of Business and Law, London Metropolitan University
gmn0021@my.londonmet.ac.uk | ORCID 0009-0009-8594-0248

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Abstract

Multilateral climate funds (MCFs) are expected to move capital at scale into the Group of 77 (G77) and China, yet a large climate finance gap persists. Drawing on institutional economics and research on fragmented climate governance, this study examines how institutional collaboration, blended finance deployment, policy and regulatory alignment, and transparency shape the operational architecture of MCFs. It combines a survey of 103 sustainable finance specialists with eight semi-structured interviews with practitioners at multilateral development banks (MDBs) and climate funds, UN bodies, a central bank and a private capital firm. Interviewees saw blended finance as the main route to mobilising private capital, but described it as held back by fragmented governance, silos between vertical funds and MDBs, and regulatory dissonance. In the survey, institutional collaboration was the only factor with a significant, independent association with architecture effectiveness. Interviewees treated policy alignment as a precondition for absorbing capital, and transparency as the basis of investor trust. The study develops an Institutional Reinforcement Cycle, which explains why these determinants act as complements, so that weakness in one of the four can stall the whole system. It concludes that MCFs and MDBs need coordinated platforms that de-risk private investment at scale.

Keywords: Climate finance; multilateral climate funds; blended finance; institutional collaboration; institutional economics; multilateral development banks; central banks; G77 and China

1. Introduction

Climate change is reshaping the global financial system. The international community has repeatedly called for sustainable and inclusive financial systems, from the Paris Agreement under the United Nations Framework Convention on Climate Change (UNFCCC) and the Sustainable Development Goals in 2015 to the Belém Political Package agreed at the 30th UNFCCC Conference of the Parties (COP30) in 2025. Central banks, private capital and multilateral development banks (MDBs) drive this shift, because they can create incentives for green finance within the financial sector (Panagopoulos and Tzionas, 2023). How these institutions across the Group of 77 (G77) and China can contribute most effectively is still not well understood, particularly in countries that are highly vulnerable to climate change, heavily indebted and exposed to acute climate-related and economic shocks. These institutions form an interdependent system, in which institutional, financial and regulatory components strengthen

or constrain one another's capacity to mobilise capital. Section 4.3 develops this idea as the Institutional Reinforcement Cycle.

Despite the growing emphasis on multilateral climate funds (MCFs), the literature on their institutional architecture remains fragmented and unevenly spread across the major funds. As a result, the interaction between institutional collaboration, blended finance deployment, regulatory alignment and transparency has not been theorised as a single, coherent system. This study addresses that gap by examining how the four determinants jointly shape the operational architecture of climate finance across the G77 and China.

The study is guided by one research question, which asks what influence institutional collaboration, the effectiveness of blended finance deployment, policy and regulatory alignment, and the transparency and accountability of fund utilisation have on the operational architecture of MCFs across the G77 and China. Its aim is to identify the roles of MCFs in shaping the architecture of climate finance and to provide recommendations to countries and MDBs, with particular attention to how collaboration between MDBs, private capital and central banks influences fund utilisation, effectiveness and governance. Four objectives follow from this aim, one for each determinant, each assessing its influence on the operational architecture of MCFs across the G77 and China.

In addressing these objectives, the study contributes the Institutional Reinforcement Cycle, a recursive model that theorises how institutional, financial and regulatory variables interact to determine climate finance effectiveness and that offers both a conceptual and a practical framework for MCFs and MDBs. The model is grounded in institutional economics, in research on fragmented climate governance and in the economics of complementary inputs. Section 2 reviews the literature, previous empirical studies and the theoretical framework, Section 3 sets out the research methodology, Section 4 presents and discusses the findings, and Section 5 concludes with policy recommendations and directions for future research.

2. Literature review

2.1 The evolving architecture of climate finance

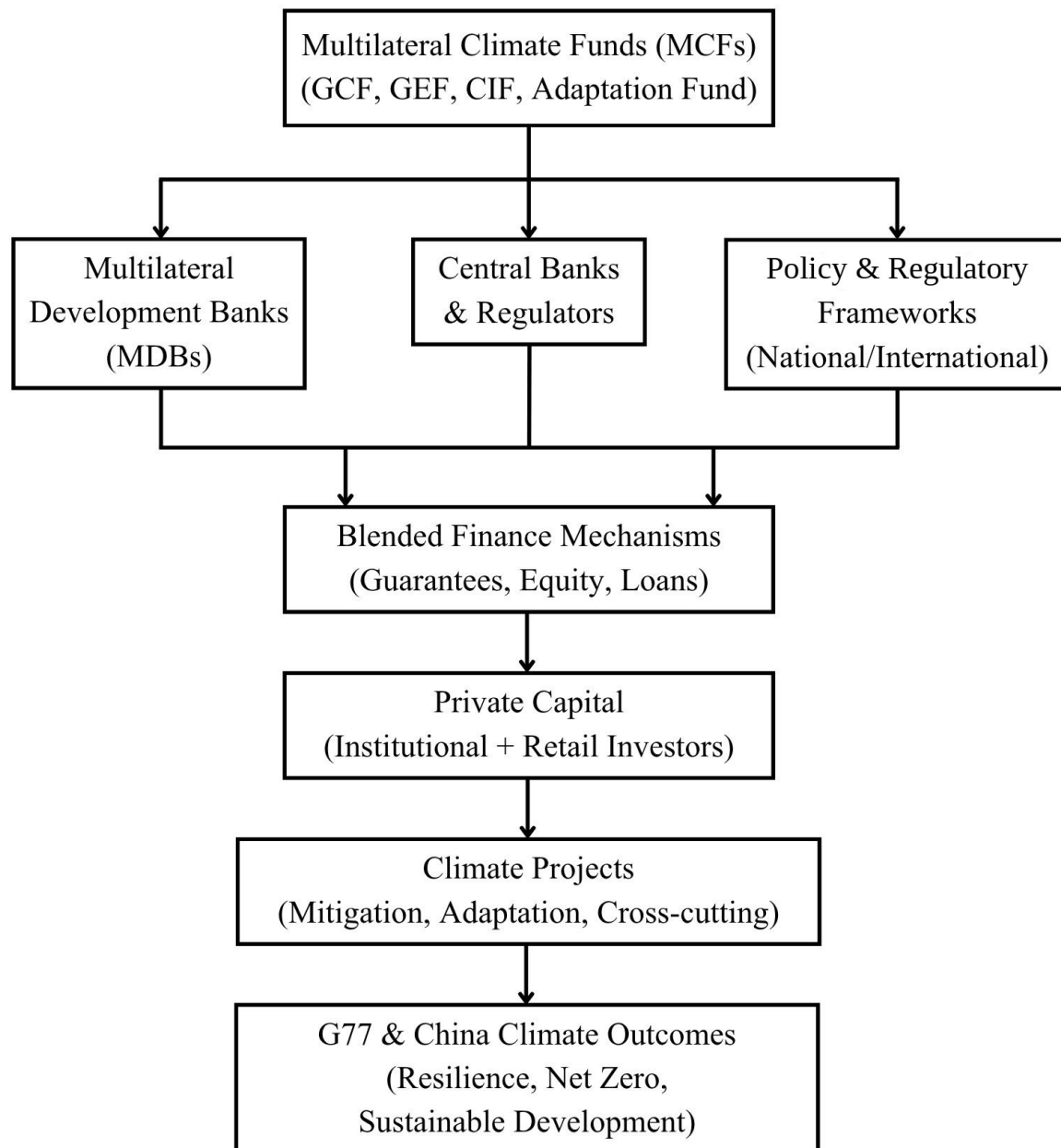
The architecture of climate finance is changing in fundamental ways. The international aid system established by the United Nations and developed countries rested on the post-war logic of Official Development Assistance (ODA), which targeted poverty reduction through financial transfers from the “rich North”, Europe and its closest allies, to the “poor South” across Asia and Africa, where many countries were still recovering from economic repression

(Orliange, 2020). The adoption of the 2030 Agenda for Sustainable Development, the Addis Ababa Action Agenda (AAAA) and the Paris Agreement at UNFCCC COP21 in 2015 marked a move away from this late twentieth-century aid architecture towards a broader framework of Global Public Goods (GPGs), placing climate change mitigation, adaptation and disaster risk reduction at the centre of international economic policy. Over the past two decades, the estimated climate finance gap has grown from billions to trillions of US dollars. The original climate finance target was USD 100 billion per year, whereas Emerging Markets and Developing Economies (EMDEs) other than China are now estimated to need around USD 1 trillion per year by 2025 and around USD 2.4 trillion per year by 2030 to meet their climate and development goals (Songwe et al., 2022). The discourse has consequently shifted towards blended finance and the role of private investment (Griffith-Jones et al., 2009), although the “billions to trillions” agenda has yet to materialise fully across the G77 and China. Multilateralism itself faces threats from geopolitical fragmentation, rising nationalism and a crisis of legitimacy in the institutions of international cooperation (Linn, 2018), and the architecture governing these flows is under severe stress as a result. The loss of trust is sharpened by the perception in the Global South that the current International Financial Architecture (IFA) is “outdated, dysfunctional, and unfair” and holds developed nations less accountable than it should (Passarelli and Justino, 2024). The available mechanisms, from the Green Climate Fund (GCF), the Global Environment Facility (GEF) and the Climate Investment Funds (CIF) to the MDBs, are increasingly seen as fragmented and bureaucratically burdensome, imposing high transaction costs on recipient countries when funds are deployed (Seballos and Kreft, 2011).

In this context, MCFs do more than transfer resources. They sit at the centre of a complex system of institutional collaboration (Kasdan et al., 2021). The literature suggests that their effectiveness depends on their capitalisation and equally on their ability to shape the operational mandates of MDBs and central banks, so that climate risk is mainstreamed into national macro-fiscal frameworks (Panagopoulos and Tzionas, 2023). The channels through which MCFs distribute funds to projects, such as concessional loans, equity and guarantees, can be described as financial and operational deployment mechanisms (Porter et al., 2008). In the G77 and China, capital availability alone is insufficient, and the choice of financial instrument and the efficiency of the deployment framework determine the eventual impact on implementation (Shipalana, 2023). Blended finance is central to modern deployment. It uses limited public or multilateral funds to de-risk projects and so mobilises private capital that

would otherwise stay on the sidelines because of perceived risk in emerging markets (Dasgupta et al., 2019). Research on China's green financial system shows how structured policy frameworks, including green initial public offering (IPO) channels and compulsory environmental insurance, can integrate these mechanisms into the wider national economy and scale their impact on the ground (People's Bank of China and UNEP, 2015).

A multi-stakeholder framework that connects public policy with private capital is therefore needed, and carbon accounting standards could help close the trillion-dollar gap in climate investment identified in the literature (Ogunyemi, 2020). Without such alignment, deployment will remain fragmented and held back by high transaction costs and a lack of standardised reporting, which further limits the ability of Small Island Developing States (SIDS) and other vulnerable G77 members to access these funds (Bishop et al., 2021). Figure 1 sets out a model of the architecture of climate finance.



*Figure 1. Climate finance architecture, showing institutional interactions and capital flow mechanisms.
Author's own.*

The literature reviewed shows that the architecture of climate finance has become a multi-layered system. MCFs connect high-level international commitments with implementation on the ground, and the architecture relies heavily on their operational leadership to bridge the persistent gap between financial markets and environmental, social and governance (ESG) goals. Success in the G77 and China depends less on the volume of capital than on how well institutional collaboration, policy alignment and the deployment of blended finance work together. These elements form the foundation of a financial architecture that seeks to balance economic growth with urgent environmental needs (Shipalana, 2023). The GCF, the GEF, the Adaptation Fund and the CIF are the four primary multilateral funds in this architecture, yet

research coverage of them is markedly uneven. Figure 2 maps the distribution of academic coverage and the research gaps that remain.

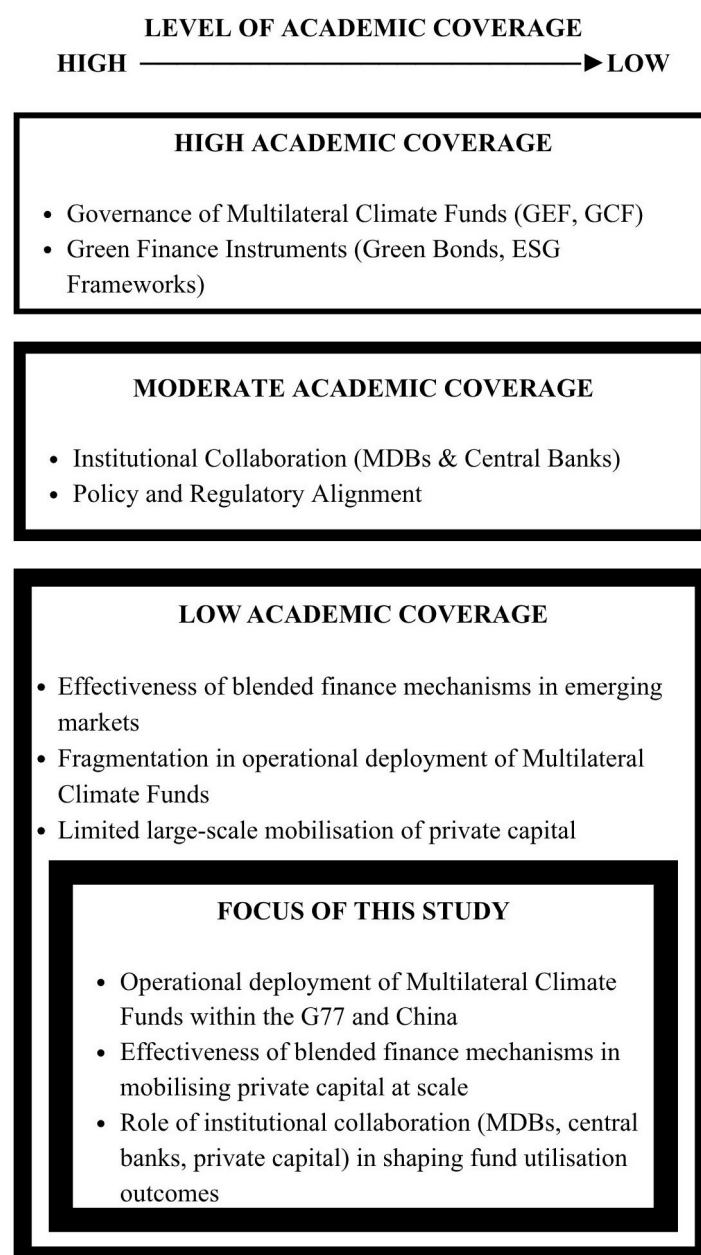


Figure 2. Literature gap and theoretical positioning map. Author's own.

2.2 Previous empirical studies of collaboration and access

A growing body of empirical work examines how multilateral funds and development banks work together. Seballos and Kreft (2011) show, for the Pilot Program for Climate Resilience, how the political economy of a multilateral programme shapes its operation at country level, and Kasdan et al. (2021) trace how multilateral funds dedicated to adaptation have come to interpret transformational change. Cofinancing has received particular attention. Kotchen and Negi (2019) examine cofinancing across 3,269 GEF projects, and Kotchen and Vogt (2025)

find, using GEF and GCF data, that adding cofinancing objectives to a fund that already seeks to maximise environmental impact is redundant at best and more likely counterproductive, with little effect on realised environmental benefits. For the present study this is an important result, since it suggests that the volume of capital mobilised is a poor guide to the effectiveness of the architecture.

More recent work addresses collaboration directly. A study commissioned by the G20 Sustainable Finance Working Group finds that collaboration between vertical climate and environmental funds, MDBs and national development banks is held back by complex accreditation, slow approvals, duplicated standards and information asymmetries, which together impose substantial transaction costs on private investors (Abraham et al., 2025). The same study finds that co-financing and mobilisation linked to these funds are concentrated in middle-income countries, which received 43% of volumes compared with 8% for low-income countries. At the national level, Anjanappa (2025) models India's climate finance system, identifying barriers to collaboration between vertical funds, MDBs, national development banks and private financial institutions, and feedback loops that can either trap investment or accelerate it.

These studies establish that coordination problems are widespread, but most examine a single fund, instrument or country, or rely on institutional data rather than practitioner accounts. Fewer bring together the views of MDB, central bank and private capital practitioners across the G77 and China to examine how the four determinants interact. The present study addresses that gap.

2.3 Theoretical framework

The study draws on three bodies of theory to explain why the four determinants might interact rather than act independently. The first is institutional economics. North (1990) treats institutions as the rules of the game that structure economic exchange and argues that institutional change is path dependent, so that early arrangements shape later options. Transaction cost economics adds that organisational arrangements matter because they determine the cost of coordinating exchange (Williamson, 1985). From this perspective, the fragmentation of MCFs, MDBs and domestic regulators is more than an administrative inconvenience. It raises the cost of every transaction that must pass between them, which is consistent with the high transaction costs reported for recipient countries (Seballos and Kreft, 2011).

The second is the literature on fragmented global environmental governance. Biermann et al. (2009) describe governance architectures as more or less fragmented, distinguishing synergistic, cooperative and conflictive fragmentation, and argue that high degrees of fragmentation are likely to reduce the overall performance of an architecture. Keohane and Victor (2011) characterise climate governance as a regime complex, a loosely coupled set of institutions rather than a single integrated regime, while Ostrom (2010) shows that polycentric systems with many centres of decision-making can perform well when those centres coordinate. Together these accounts suggest that the performance of climate finance depends less on any single institution than on the quality of coordination between institutions, which is the role that institutional collaboration plays in this study.

The third is the economics of complementary inputs. Kremer's (1993) O-ring theory describes production processes in which a failure in one of several tasks destroys much of the value of the whole, so that quantity cannot substitute for quality and small weaknesses have large effects. Applied to climate finance, the analogy suggests that capital cannot compensate for weak collaboration, misaligned regulation or poor transparency, because each is needed for capital to be deployed effectively. Pierson (2000) adds a temporal dimension, showing how increasing returns make institutional processes self-reinforcing, so that systems which start well tend to improve while those which start badly tend to remain locked in.

These three strands underpin the Institutional Reinforcement Cycle developed in Section 4.3. Institutional economics explains why coordination costs matter, the governance literature explains why fragmentation weakens the architecture as a whole, and the theories of complementary inputs and increasing returns explain why the four determinants are expected to interact multiplicatively and to reinforce one another over time. The framework therefore leads to two expectations that the interviews and survey are used to examine. The first is that institutional collaboration will be more closely associated with the effectiveness of the architecture than the volume of blended finance. The second is that weakness in a single determinant will limit the contribution of the others.

2.4 Conceptual framework

The conceptual framework rests on four interconnected pillars. These are institutional collaboration, the effectiveness of blended finance deployment, policy and regulatory alignment, and transparency. As Figure 3 shows, it builds on the theories set out above by emphasising the distinct role of MCFs in driving sustainable practice across emerging markets.

It treats institutional collaboration between MDBs and central banks as the primary driver of the circulation of finance, while the choice of deployment mechanism shapes how useful the funds are to recipient countries. The review also indicates that a fair international financial architecture requires more than policy commitments. It requires a fundamental change in how funds are operationalised to meet the specific needs of countries within the G77 and China. By linking green central banking with the operational mandates of the four major MCFs, the framework locates the capacity of climate finance to drive change in its ability to align corporate and national strategies with verifiable environmental outcomes through transparency. This provides the basis for examining how these funds can move beyond the role of aid provider and become central actors in the global financial system (Shirai, 2023).

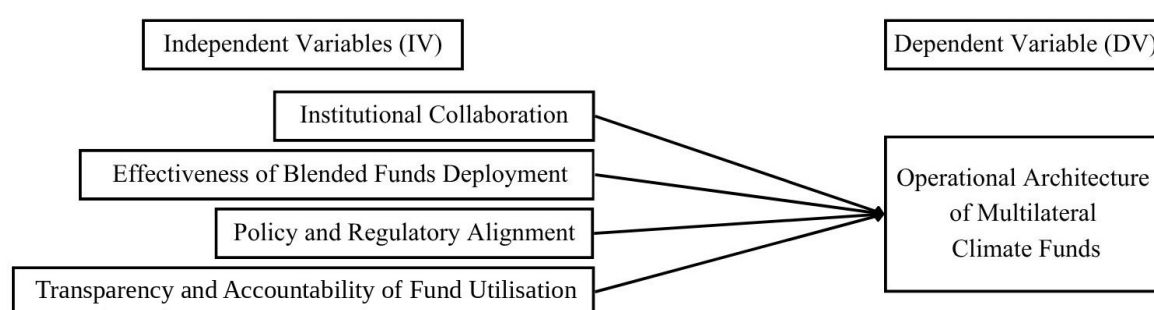


Figure 3. Conceptual framework. Author's own.

3. Research methodology

The study used a mixed-methods design. It combined eight semi-structured interviews with a cross-sectional survey of sustainable finance specialists, and the two strands were used to triangulate one another (Creswell, 1998). The interviewees were practitioners at MDBs and climate funds (the African Development Bank, the Asian Development Bank and the World Bank GEF), UN bodies (UNDP/UNCDF, the UN Sustainable Stock Exchanges initiative and the Food and Agriculture Organization), a central bank and a private capital firm. The research strategy examined a contemporary phenomenon within its real-life institutional context, since the boundaries between the phenomenon and that context are not easily separated (Yin, 2018). Data were collected at a single point in time across more than one case, which allowed patterns across institutional, financial and regulatory variables to be identified and compared (Bryman, 2012).

The qualitative strand explored the reasoning behind institutional decision-making, using descriptive data drawn from practitioner accounts (Creswell, 1998). Individual interviews were chosen over focus group discussions because they allow deeper analysis of individual professional insight and fit better around the demanding schedules of senior executives. Each

interview lasted between 45 and 60 minutes and was recorded with the participant's prior consent.

The survey used a structured questionnaire, written in English and distributed online through Microsoft Forms, to capture a broader picture of practitioner views across the G77 and China alongside the interview data. A sample of 103 was drawn from a population of 144 sustainable finance professionals working in MDBs, central banks, private capital and climate funds, using standard sample-size determination tables for a known population. Internal consistency was assessed with Cronbach's alpha against a target coefficient of 0.70 or higher, and content validity was supported by the literature review and expert consultation. Table 1 shows the distribution of the sample by sector and region, and Table 2 sets out the survey constructs.

Table 1. Survey sample distribution by sector and region. Author's own.

Sector or stakeholder group	G77 nations	China	Total
MDBs	35	10	45
Central banks and regulatory bodies	22	8	30
Private capital stakeholders and ESG leads	20	8	28
Total	77	26	103

Table 2. Survey constructs and associated research propositions. Author's own.

Construct	Research proposition
Institutional collaboration (IV1)	Institutional collaboration is associated with the operational architecture of multilateral climate funds across the G77 and China.
Blended finance deployment (IV2)	The effectiveness of blended finance deployment is associated with the operational architecture of multilateral climate funds across the G77 and China.
Policy and regulatory alignment (IV3)	Policy and regulatory alignment is associated with the operational architecture of multilateral climate funds across the G77 and China.
Transparency and accountability (IV4)	Transparency and accountability of fund utilisation is associated with the operational architecture of multilateral climate funds across the G77 and China.

Each strand was analysed with techniques suited to it. Interview transcripts were analysed through thematic content analysis, which identified recurring themes and the factors affecting the effectiveness and transparency of climate fund utilisation. The resulting codes were grouped into three themes and mapped to the four determinants and the dependent variable (Table 3). Survey responses were analysed descriptively in IBM SPSS Statistics, and Pearson correlation and multiple regression were used to explore the strength and independence of associations between the four constructs and the operational architecture of MCFs. The survey

analysis provides quantitative context for the interview findings and is not intended as a standalone test of the conceptual framework.

4. Findings and analysis

4.1 Analysis of interviews

The qualitative strand consisted of eight semi-structured interviews with senior practitioners at MDBs and climate funds, UN bodies, a central bank and a private capital firm. Content analysis of the transcripts produced three dominant themes. These were fragmented governance and institutional silos, the catalytic limits of blended finance, and regulatory dissonance in emerging markets. Table 3 summarises them.

Table 3. Thematic matrix of qualitative findings and variable mapping. Author's own.

Dominant theme	Operational concepts and codes	Mapped research variables
Theme 1. Fragmented governance and institutional silos	Vertical fund and MDB friction; bureaucratic inertia; strategic alignment failure; duplication of initiatives; high transaction costs	IV1 Institutional collaboration; IV4 Transparency and accountability; DV Climate finance architecture
Theme 2. The catalytic limits of blended finance	Operational friction in deployment; concessional finance inefficiency; time-lag disincentive; private capital mobility; over-centralisation of approvals	IV2 Effectiveness of blended finance; IV4 Transparency and accountability; DV Climate finance architecture
Theme 3. Regulatory dissonance in emerging markets	Domestic policy preconditions	IV3 Policy and regulatory alignment; DV Climate finance architecture

Under the first theme, fragmented governance and institutional silos, interviewees broadly agreed that although MCFs have the mandate to drive transformational change, their operational design often conflicts with the bureaucratic inertia of traditional MDBs. One MDB official described the architecture as entirely fragmented. In the official's words, "We have vertical funds demanding rapid deployment for adaptation, but our internal risk committees are still functioning under post-WWII creditworthiness metrics." Participants frequently said that institutional collaboration is treated as a compliance exercise rather than a strategic priority. The analysis also showed that the absence of joint decision-making mechanisms often leads to duplicated climate finance initiatives, which raises transaction costs for recipient countries in the G77.

Under the second theme, the catalytic limits of blended finance, all eight participants accepted that blended finance is, in principle, the most viable way to mobilise private capital. They also

described considerable operational friction. As one private capital stakeholder explained, “Concessional finance and first-loss guarantees look excellent on paper, but the timeline to secure these from an MCF such as the GCF takes years, by which point the private capital has moved to more liquid, traditional assets.” Interviewees added that scaling up would require MCFs to give regional MDBs more operational autonomy instead of centralising project approvals.

The third theme, regulatory dissonance in emerging markets, concerned policy alignment and central banking. Several experts stressed that MCFs cannot succeed without a supportive domestic regulatory environment. A central bank policy adviser observed that “Climate finance cannot just be an external aid mechanism, it must be integrated into our macro-fiscal frameworks. Unless central banks in the G77 actively adjust prudential policies to favour green assets, MCF funding will remain isolated pilot projects.” These accounts indicate that regulatory alignment is a fundamental requirement for the demand, deployment and implementation of climate finance across the G77 and China, rather than a supplementary variable.

Across the three themes, interviewees described problems of coordination more often than problems of capital availability. Blended finance was valued, but its effect was seen to depend on whether collaboration between institutions, approval processes and domestic regulation allowed it to be used. Section 4.2 sets this pattern against the survey evidence, and Section 4.3 interprets it in the light of the theoretical framework.

4.2 Survey analysis

Before inferential tests were run, the internal consistency of the instrument was assessed by calculating Cronbach’s alpha for the item sets of the four independent variables (IV1 to IV4) and the dependent variable (DV). Coefficients ranged from 0.813 to 0.863, above the conventional 0.70 threshold.

Descriptive analysis, based on a five-point Likert scale and summarised with measures of central tendency and dispersion, gave a baseline picture of respondent views on each construct. For institutional collaboration (IV1), respondents agreed that partnerships between MCFs and private capital providers improve access to finance (Mean = 3.89, standard deviation (SD) = 0.766) and supported central bank engagement (Mean = 3.96). Agreement was strongest for the statement that collaboration reduces the duplication of climate finance initiatives (Mean = 4.52, SD = 0.654). For blended finance deployment (IV2), there was broad agreement on the need for

de-risking instruments, with statements on risk-sharing mechanisms and on the ability of blended structures to mobilise private capital yielding means of 3.84 and 3.82 respectively. Standard deviations of 0.683 and 0.678 point to some variation in respondents' experience. Statements on operational design and on co-financing arrangements between MDBs and private capital scored lower (3.33 and 3.35), which suggests operational friction in deploying instruments that respondents regard as sound in principle. Policy and regulatory alignment (IV3) produced some of the highest individual scores alongside notable variance. The statement that clear regulatory standards improve transparency had one of the highest means in the survey (Mean = 4.01, mode = 4), whereas the statement that current regulatory frameworks in G77 countries facilitate integration scored considerably lower (Mean = 3.23, SD = 0.992). For transparency and accountability (IV4), statements on transparent reporting mechanisms and on transparency in fund utilisation recorded the two highest means in the construct (3.81 and 3.85). Every item in this construct had a mode of 4, and item means fell within a narrow band from 3.66 to 3.85, the most uniform pattern of any construct. The pattern suggests that finance professionals see standardised impact measurement and accountability as a prerequisite for building stakeholder trust and mobilising private capital, more than as a matter of ethical compliance. Perceptions of the current climate finance architecture, the dependent variable, were largely neutral to pessimistic, in contrast to the high scores given to the independent constructs. The statement that the current institutional structure of MCFs promotes coherence between public and private climate finance actors recorded one of the lowest means in the survey (Mean = 3.44, SD = 0.825), and views on the coherence of the current institutional design were similarly weak (Mean = 3.46).

Pearson correlation and multiple regression were used to explore associations between the four constructs and the operational architecture of MCFs (Table 4). Institutional collaboration showed a moderate, positive and significant correlation ($r = .409$, $p < .001$) and was the only construct to retain a significant, independent association in the regression ($\beta = .390$, $p < .001$), uniquely accounting for approximately 14% of the variance in the dependent variable within a model that explained 20.7% overall (adjusted $R^2 = .174$). Blended finance deployment was significantly associated at the bivariate level ($r = .214$, $p = .030$), but the association did not hold once the other constructs were accounted for in the regression ($\beta = .121$, $p = .196$). Policy and regulatory alignment showed no significant association at either the bivariate level ($r = -.005$, $p = .958$) or the regression level ($\beta = .024$, $p = .793$). Transparency showed a positive but non-significant association at both levels ($r = .150$, $p = .132$, and $\beta = .145$, $p = .111$). On

these quantitative measures, institutional collaboration is the construct most consistently associated with architecture effectiveness, while the other three constructs are better understood through the interview findings discussed below.

Table 4. Pearson correlations and multiple regression results for the four constructs and the operational architecture of MCFs. Author's own.

Construct	Pearson r	p	Beta	p
Institutional collaboration (IV1)	.409	< .001	.390	< .001
Blended finance deployment (IV2)	.214	.030	.121	.196
Policy and regulatory alignment (IV3)	-.005	.958	.024	.793
Transparency and accountability (IV4)	.150	.132	.145	.111

Note. N = 103. Dependent variable is the perceived operational architecture of MCFs. Beta is the standardised regression coefficient. $R^2 = .207$, adjusted $R^2 = .174$.

4.3 Discussion of findings

The survey found a significant, positive relationship between institutional collaboration and the effectiveness of the climate finance architecture. Interviewees frequently noted that MDBs and vertical funds such as the GEF and the GCF operate with overlapping mandates but different deployment criteria. This fragmentation reduces the agility of the funds. Institutional economics would predict exactly this pattern. Where MCFs, MDBs and regulators each apply their own criteria, every project bears the cost of reconciling them (Williamson, 1985), and the architecture takes on the more fragmented forms that Biermann et al. (2009) associate with weaker overall performance. The study adds to existing knowledge by showing how this collaboration needs to evolve. The survey data and interview insights together suggest that MDBs should move from acting mainly as lenders to acting as catalytic platforms. In the interviewees' account, institutional collaboration becomes effective only when MCFs delegate project-level risk assessment to regional MDBs, which can draw on local market intelligence, instead of centralising decisions. The G20-commissioned study by Abraham et al. (2025) reaches a similar conclusion from institutional data, recommending that funds delegate project-by-project approvals to accredited entities, which is close to the delegation to regional MDBs that interviewees proposed. Such delegation runs against traditional, highly centralised governance models and is consistent with the emergence of South-South institutional alternatives, such as the Asian Infrastructure Investment Bank (AIIB) and the New Development Bank, through which borrowing countries seek a seat at the table rather than simply receiving aid.

Blended finance dominated the interviews, although its statistical association with architecture effectiveness did not hold once institutional collaboration was controlled for ($\beta = .121$, $p = .196$). Interviewees nonetheless described concessional loans, guarantees and risk-sharing instruments as the primary drivers of crowding in private capital in the G77 and China. They stated explicitly that without MCFs acting as a first-loss buffer, private equity and institutional investors regard the macroeconomic risks in G77 countries as prohibitive. The interviews also revealed a severe timeline mismatch, since MCFs work too slow, bureaucratic public-sector timelines while private capital demands rapid and efficient deployment (Shipalana, 2023). The more immediate constraint is therefore operational inefficiency rather than a shortage of capital. The loss of statistical significance once collaboration is accounted for is consistent with Kotchen and Vogt (2025), who find that an emphasis on cofinancing adds little to realised environmental benefits. The volume of capital mobilised appears to depend on the coordination that surrounds it. These findings are consistent with the systematic review by Ramos Farroñán et al. (2025), which shows that in emerging economies such as those of the G77 and China, operational design and instrument selection directly determine investment capacity. To support a functional architecture, MCFs need to standardise their blended finance offerings into predictable, off-the-shelf risk-sharing products instead of treating every project as a bespoke negotiation lasting several years.

Policy and regulatory alignment was not significantly associated with the dependent variable in this sample ($r = -.005$, $p = .958$, and $\beta = .024$, $p = .793$), yet interviewees identified it as a fundamental influence on how climate finance moves across the G77 and China. Survey responses on regulatory integration also showed a high standard deviation ($SD = 0.992$), which points to wide variation in the regulatory contexts in which respondents work across the G77 and China. This pattern is consistent with China's experience, where green finance has developed within a highly structured national policy framework (People's Bank of China and UNEP, 2015). China sits at the successful end of this spectrum, with national policy closely aligned with green finance goals. By contrast, interviewees described regulatory dissonance in many other G77 countries, including Nepal, Lesotho, Rwanda, Bhutan and Ethiopia. The findings extend Shirai's (2023) discussion of green central banking, since interviewees stressed that central banks must become proactive agents in the climate finance architecture by integrating climate risks into prudential policy. Where a domestic central bank does not incentivise green lending, for example through favourable reserve requirements for green assets, capital deployed by MCFs meets a domestic bottleneck that domestic political realities

make difficult to overcome. In Ostrom's (2010) terms the architecture is polycentric, but its domestic centres of decision-making are often poorly coordinated with its international ones. The findings also support the move towards a Multidimensional Vulnerability Index (MVI), so that policy alignment translates into actual concessional access for the most at-risk countries.

Transparency did not reach statistical significance in the survey ($r = .150$, $p = .132$), yet interviewees described it as the currency of trust in the climate finance architecture, and private capital providers across the Middle East, developed economies and emerging economies treated clear impact measurement standards as essential. This is consistent with Gilchrist et al. (2021), who warn that poorly implemented deployment frameworks and weak governance reduce transparency and limit the credibility of green finance instruments. In the interviews, greenwashing emerged as a major deterrent for private investors. Institutional investors are wary of deploying capital alongside MCFs in G77 countries without standardised, universally accepted greenhouse gas (GHG) accounting protocols. Transparent reporting is therefore a financial prerequisite for scale as well as an ethical requirement.

Together, the four pillars of the conceptual framework give an integrated evaluation of the MCF system. The findings bear out the problem set out at the start of the paper, in which the move from profit-maximising approaches to integrated environmental and social approaches is held back by inefficiencies in the architecture. They also confirm that the G77's perception of the International Financial Architecture as outdated, dysfunctional and unfair is rooted in operational realities. The study therefore proposes the conceptual contribution illustrated in Figure 4, an Institutional Reinforcement Cycle in which institutional collaboration, policy and regulatory alignment, blended finance effectiveness and transparency reinforce one another over time, while a single weak link can stall the whole cycle. The cycle combines the mechanisms set out in Section 2.3. From the theory of complementary inputs it takes the idea that the four determinants interact multiplicatively, so that if one of them approaches zero the reinforcing effect of the whole cycle collapses and additional capital cannot make up the difference (Kremer, 1993). From the literature on increasing returns it takes the idea that effectiveness carries over from one period to the next, so that well-coordinated systems tend to improve while poorly coordinated ones remain locked in (Pierson, 2000). Figure 4 labels the three parameters that govern this process, namely the rate at which institutional gains depreciate, the catalytic multiplier and the elasticity of synergy between the determinants. The cycle explains why climate finance effectiveness varies so widely across the G77 and China, since similar volumes of capital can produce very different outcomes depending on the weakest

link in each system. The multiplicative structure is shown in Figure 5, and a formal mathematical statement of the cycle is provided in Appendix A.

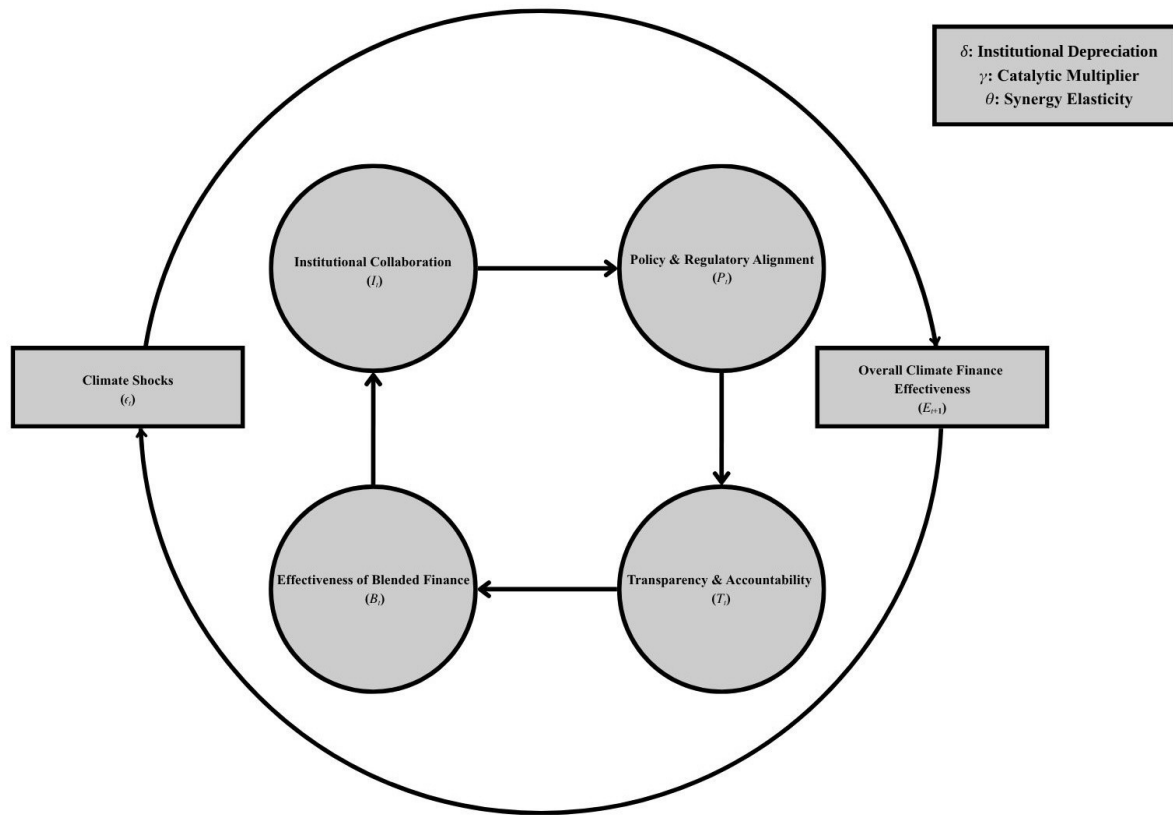


Figure 4. The Institutional Reinforcement Cycle of multilateral climate finance. Author's own.

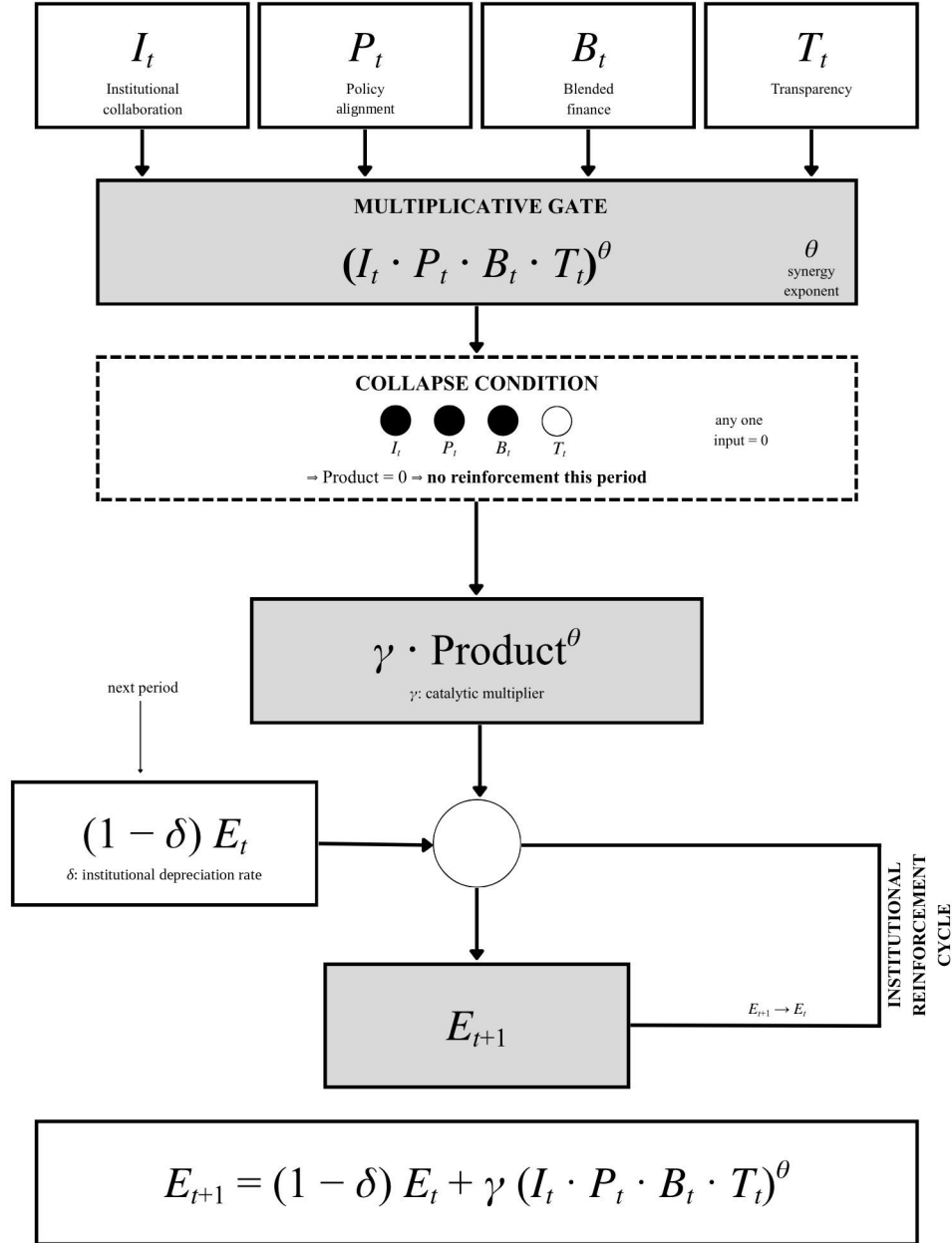


Figure 5. The Institutional Reinforcement Cycle as a multiplicative gate with memory. Author's own.

By treating climate finance as a self-reinforcing, dynamic system rather than a set of isolated linear interventions, the study offers a more rigorous explanation of how MCFs need to operate to achieve scale. Moving beyond traditional ODA requires MCFs to restructure themselves. The findings show that capital volume matters less than deployment efficiency, and that the operational architecture will succeed in raising the trillions required only if central banks integrate climate risks and MCFs use MDBs as local catalytic deployment channels rather than administrative hurdles and standardise blended finance to de-risk private capital quickly.

5. Conclusions

The research question asked how institutional collaboration, blended finance deployment, regulatory alignment and transparency influence the operational architecture of climate finance. Only institutional collaboration retained a significant, independent association in the regression, while blended finance, regulatory alignment and transparency emerged from the interviews as perceived determinants of the architecture across the G77 and China. The interviews indicate that blended finance is seen as the primary catalyst for mobilising private capital, and that its success depends on institutional collaboration to remove bureaucratic silos and on domestic regulatory alignment to ensure the capital can be absorbed. Transparency and accountability are seen as the trust-building mechanisms that sustain private capital participation. These variables do not operate in isolation. They form an interdependent Institutional Reinforcement Cycle that governs how effectively MCFs can close the global investment gap. Theoretically, the study links institutional economics, research on fragmented climate governance and the economics of complementary inputs, and uses them to explain why the quality of coordination, rather than the volume of capital, shapes the effectiveness of MCFs.

Four recommendations follow. First, institutional collaboration should be prioritised, with MCFs acting as coordination platforms that actively link MDBs, central banks and private capital providers. Joint financing platforms and shared risk-assessment matrices would reduce transaction costs and make deployment more efficient across developing economies. Second, policymakers should give urgent attention to policy and regulatory alignment in order to activate the multiplier within the reinforcement cycle. Governments in the G77 and China should develop unified green finance taxonomies, enforce mandatory ESG disclosure requirements and provide long-term policy stability, and central banks should integrate climate-related financial risks into their monetary and supervisory frameworks, treating climate finance as a core financial-stability priority rather than a peripheral environmental objective. Third, the deployment of blended finance should be scaled through stronger risk-sharing structures, with MCFs expanding the use of first-loss guarantees, concessional loans and targeted co-financing arrangements to de-risk private capital entry, particularly in high-risk, climate-vulnerable SIDS and Least Developed Countries (LDCs). Finally, transparency and accountability should be strengthened. Standardised performance indicators and rigorous, verifiable impact reporting would build the investor trust needed to sustain the cycle and secure a scalable climate finance architecture.

The study has limitations. The quantitative sample of 103 sustainable finance specialists and the qualitative sample of eight semi-structured interviews, while appropriate for the mixed-methods design, constrain the generalisability of the findings and may under-represent the perspectives of smaller or less-resourced institutions across the G77 and China. The cross-sectional design captures perceptions of the climate finance architecture at a single point in time, and reliance on self-reported data from institutional representatives may introduce social desirability or affiliation bias.

Future research should examine how the Institutional Reinforcement Cycle operates within developed economies. Longitudinal studies are needed to track how institutional collaboration and policy alignment evolve and reinforce one another over time. Future work should also evaluate empirically which blended finance instruments, for example green bonds or policy-based guarantees, have the greatest catalytic effect across different regulatory environments in the Global North and the Global South, with particular attention to the transfer of capital from developed to developing economies.

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Appendix A. Formal specification of the Institutional Reinforcement Cycle

A.1 Purpose

This appendix sets out the formal statement of the Institutional Reinforcement Cycle (IRC) described in Section 4.3 and shown in Figures 4 and 5. The specification formalises the relationships proposed in this paper so that they can be calibrated and tested in future work. It is a conceptual model and has not been estimated in this study.

A.2 Variables and parameters

E_t denotes overall climate finance effectiveness in period t . The four determinants are institutional collaboration (I_t), policy and regulatory alignment (P_t), the effectiveness of blended finance deployment (B_t) and the transparency and accountability of fund utilisation (T_t). Each determinant is non-negative and measured on a common normalised scale.

Three parameters govern the cycle. The institutional depreciation rate δ , with $0 < \delta \leq 1$, is the share of effectiveness lost in each period in the absence of reinforcement. The catalytic multiplier $\gamma > 0$ scales the reinforcement produced by the four determinants. The synergy elasticity $\theta > 0$ governs how strongly the determinants reinforce one another.

A.3 The model

Effectiveness evolves according to

$$E_{t+1} = (1 - \delta) E_t + \gamma (I_t \cdot P_t \cdot B_t \cdot T_t)^\theta \quad (1)$$

It is convenient to write the reinforcement term as

$$R_t = \gamma (I_t \cdot P_t \cdot B_t \cdot T_t)^\theta \quad (2)$$

so that $E_{t+1} = (1 - \delta) E_t + R_t$. The first term carries effectiveness forward from one period to the next, which gives the cycle its memory. The second term is the multiplicative gate through which the four determinants add to effectiveness.

A.4 Properties

Five properties follow directly from equations (1) and (2).

Collapse condition: If any of I_t , P_t , B_t or T_t equals zero, then $R_t = 0$ and $E_{t+1} = (1 - \delta) E_t$. Effectiveness then only decays, and no increase in the other determinants can offset the missing one.

Complementarity: For any two determinants X and Y with strictly positive values, the cross-partial derivative of R_t is $\theta^2 R_t / (XY) > 0$. An improvement in one determinant therefore raises the marginal contribution of every other determinant.

Weakest-link effect: The marginal contribution of determinant X is $\theta R_t / X$. It is largest where that determinant is weakest, while the elasticity of R_t with respect to each determinant is the same, equal to θ . Improving the weakest determinant therefore yields the greatest absolute gain in reinforcement.

Joint improvement: R_t is homogeneous of degree 4θ in the four determinants. A proportional improvement in all four raises reinforcement more than proportionally when $\theta > 1/4$.

Memory and steady state: If the determinants are held constant, so that $R_t = R$ in every period, effectiveness converges to the steady state $E^* = R / \delta$, following the path $E_t = E^* + (1 - \delta)^t (E_0 - E^*)$. A lower depreciation rate raises the steady state and slows the adjustment towards it.

A.5 Climate shocks

Figure 4 also shows climate shocks (ε_t) acting on the cycle. These can be introduced as an additive disturbance

$$E_{t+1} = (1 - \delta) E_t + \gamma (I_t \cdot P_t \cdot B_t \cdot T_t)^\theta + \varepsilon_t \quad (3)$$

A shock then persists in effectiveness through the carry-over term, shrinking by a factor of $(1 - \delta)$ in each subsequent period.

A.6 Relation to the evidence in this paper

The evidence in this paper bears on the structure of the model rather than on its parameter values. The interviews describe institutional collaboration, regulatory alignment and transparency as preconditions for the effective use of blended finance, which is consistent with the multiplicative form, and the survey finds that institutional collaboration is the construct most consistently associated with the effectiveness of the architecture. Estimating δ , γ and θ requires panel data on the four determinants and on realised finance outcomes, which is a task for future research.